| INCH-POUND |

S-R-2834
December 30, 1993
SUPERSEDING
MIL-R-11723E
20 October 1988

FEDERAL SPECIFICATION

RADIATORS: HEATING, STEEL, MULTIFIN TYPE

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers multifin-type steel radiators, with mounting brackets and covers.

1.2 Classification. Radiators will be of the following sizes and lengths, as specified (see 6.2):

Size 1 - 1-1/4-inch (31.75 millimeter (mm)) nominal pipe size
Size 2 - 2-inch (50.8 mm) nominal pipe size

Length - 2-feet (0.61 meter (m)) to 10-feet (3.04 m) in 6-inch (152.4 mm) increments
Length - 11-feet (3.35 m)
Length - 12-feet (3.66 m)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, 1000 23rd Avenue, Port Hueneme, CA 93043-4301, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4520

AMSC N/A

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Federal Standard

FED-STD-123 - Marking for Shipment (Civil Agencies)

Military Specifications

MIL-P-116 - Preservation, Methods of
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible

Military Standards

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-2073-1 - DoD Material Procedures for Development and Application of Packaging Requirements

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents which are current on the date of the solicitation (see 6.2).

American National Standards Institute, Inc. (ANSI)

ANSI B1.20.1 - Pipe Threads, General Purpose (Inch)

(Application for copies should be addressed to the American National Standards Institute, Inc., 11 West 42nd Street, New York, NY 10036.)

ASTM

ASTM A53 - Pipe, Steel, Black and Hot-Dipped, Zinc-Coated Welded and Seamless
ASTM A178/A178M - Electric-Resistance-Welded Carbon Steel and Carbon-Manganese Steel Boiler Tubes
ASTM A192/A192M - Seamless Carbon Steel Boiler Tubes for High-Pressure Service

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ASTM A214/A214M - Electric-Resistance-Welded Carbon Steel Heat-Exchanger and Condenser Tubes
ASTM A569/A569M - Steel, Carbon (0.15 Maximum, Percent), Hot-Rolled Sheet and Strip Commercial Quality
ASTM D3951 - Commercial Packaging

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

Hydronics Institute

IBR Testing and Rating Code for Finned Tube (Commercial) Radiation

(Application for copies should be addressed to the Hydronics Institute, 35 Russo Place, Berkeley Heights, NJ 07922.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The radiator shall consist of a steel core equipped with steel fins. Unless otherwise specified (see 6.2), the radiator shall be furnished with a protective cover, mounting brackets, and fasteners.

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2.1 and 6.2).

3.3 Standard commercial product. The radiator shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the radiator being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.3.1 System of measurement. The dimensions used in this specification are not intended to preclude the use of the metric system of measurement in the fabrication and production of the material, individual parts, and the finished product, provided form, fit, and function requirements are satisfied.

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same

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quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specified.

3.5 Design. The radiator shall be designed to permit easy accessibility for maintenance and service in the field. The design shall be such as to prevent condition hazardous to personnel or deleterious to equipment.

3.5.1 Capacity. The minimum heating capacity of the radiators shall be as shown in table I. The radiators shall be rated in accordance with the IBR Testing and Rating Code for Finned Tube (Commercial) Radiation.

TABLE I. Capacity - Steam rating Btu/hr/ft (W/m) of finned length.

Bare radiators (painted but without cover)	
Radiator Core Size	Steam Rating at 214°F -- 217°F (Btu/hr) (101.11°C -- 102.78°C (W))
1	990 (Btu/hr) (290.07 W)
2	1330 (Btu/hr) (388.69 W)

3.5.2 Operating conditions. The painted radiator with cover shall not be damaged when subjected to a working pressure of 0 to 150 pound-force per square inch gage (psig) (0 to 1034.21 kilopascals (kPa)(gage)) with alternate heating and cooling from temperatures of 358 degrees Fahrenheit (°F) (181.11° Celsius (C)) to between 60 and 90°F (15.56 and 32.22°C).

3.5.3 Pressure. All elements of the radiator shall withstand 225 psig (1551.31 kPa) pressure under water for 30 minutes without leakage or loosening of the fins.

3.6 Construction. The enclosure shall enclose the entire finned length of the radiator. The enclosure shall be fabricated from sheet steel of not less than 0.0478 inch (1.21 mm) thick and shall have openings that are not larger than 0.75 square inch (483.87 square millimeters) in area. The enclosure shall not reduce the heating capacity of the radiator by more than 5 percent of the values specified herein and shall not warp when the radiator is subjected to the operating conditions specified herein. The enclosure may be in sections or as one piece. When sectional enclosure is furnished, the top and front sections shall be equipped with a fastening device.

3.6.1 Core. The core shall be fabricated from the specified size steel pipe or steel tubing (see 6.2). Pipe shall be standard weight (Schedule 40), black,

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conforming to ASTM A53. Tubing shall conform to ASTM Designation A178, A192, or A214. For pipe, the size 1 cores shall have an outside diameter (od) of 1.625 inches (41.28 mm) with a wall thickness of 0.125 inch (3.18 mm) and the size 2 cores shall have an od of 2.375 inches (60.33 mm) with a wall thickness of 0.154 inch (3.91 mm). For tubing, the core od shall be the same as pipe core od and the wall thickness shall be a minimum of 0.125 inch (3.18 mm). At each end, the core shall extend not less than 2-1/2 inches (63.50 mm) beyond the fins. The core ends shall be either threaded or beveled for welding as specified (see 6.2). Threads shall be right-hand, external, taper pipe threads in accordance with ANSI B1.20.1. Both ends of the core shall be protected with metal or heavy plastic caps.

3.6.2 Fins. Unless otherwise specified (see 6.2), the steel fins shall be rectangular. The length or width of each rectangular fin shall be not more than 5-1/4 inches (133.35 mm). When spiral steel fins are specified, the overall diameter shall be not more than 4-1/2 inches (107.95 mm). The fins shall be secured to the core and shall not loosen, warp, or buckle when the radiator is subjected to the operating conditions specified herein. The fins shall be fabricated of steel conforming to ASTM A569.

3.7 Treatment and painting. Unless otherwise specified (see 6.2), the radiator cover shall be painted and the brackets shall be treated or painted or treated and painted in accordance with the manufacturer's standard practice. The threaded surfaces and inside of the radiator core shall not be painted.

3.8 Workmanship. All parts of the radiator, including stampings, machined surfaces, and welded parts shall be clean and free from dirt, scale, flux, and other harmful extraneous material. The covers shall be free from rough or cutting edges or sharp objects.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one radiator unit when a first article is required (see 3.2 and 6.4). This inspection shall include the examination of 4.4 and the tests of 4.5. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4, the tests of 4.5.1, and the packaging inspection of 4.6. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. The unit of product shall be radiators. All radiators offered for delivery at one time shall be considered a lot for the purpose of inspection.

4.3.1 Sampling for examination. Examination shall be based on inspection level S-2 and an Acceptable Quality Level (AQL) of 4.0 percent defective for major defects and 6.5 percent defective for minor defects.

4.3.2 Sampling for tests. Tests shall be based on inspection level S-2 and an AQL of 1.0 percent defective.

4.4 Examination. Each unit selected in accordance with 4.3.1 shall be examined to verify compliance with the requirements of this specification. Examination shall be conducted as specified in table II. Any radiator in the sample containing one or more defects shall be rejected, and if the number of defective units in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. All defective units shall be replaced with non-defective units prior to testing.

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TABLE II. Classification of defects.

Categories	Defects
Major:	
101	Parts missing or not as specified.
102	Component parts not interchangeable.
103	Construction precludes easy accessibility for maintenance or service.
104	Materials not of proper quality.
105	Dimensions not as specified.
106	Extended surfaces (fins) not properly bonded to tubes.
107	External connections not adequate.
108	Tubes not properly pitched to avoid trappings.
Minor:	
201	Painting of poor grade.
202	Workmanship not of high quality.
203	Identification marking not correct.

4.5 Tests. The first article, when a first article is required, and each sample selected in accordance with 4.3.2, shall be tested. Any unit failing to pass the following tests shall be rejected, and if the number of defective units in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. Tests shall be conducted as outlined in the referenced documents as herein specified.

4.5.1 Pressure test. Subject the radiator to air pressure as specified in 3.5.3. Any evidence of leakage or separation of the fins from the core shall constitute failure of this test.

4.5.2 Capacity. Test the radiator for capacity in accordance with the IBR Testing and Rating Code for Finned Tube (Commercial) Radiation. Nonconformance to 3.5.1 shall constitute failure of this test.

4.6 Preparation for delivery inspection. The preservation, packaging, packing, and marking of the units shall be inspected to verify compliance with the requirements of section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Methods of preservation. Cleaning processes, drying procedures, preservatives, and methods of preservation specified in the following paragraphs are listed in MIL-P-116 and shall conform to the requirements of MIL-P-116 and any applicable specifications.

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5.1.1.2 Cleaning and drying. Prior to the applications of preservative compounds or paint, surfaces shall be cleaned by process C-1 and dried by any applicable procedure of MIL-P-116.

5.1.1.3 Unprotected surfaces. Unpainted ferrous metal surfaces, including threaded surfaces, shall be coated with type P-21 preservative in accordance with MIL-P-116. Care should be taken not to coat fins, coils, or other areas that would impair the heat transfer capabilities of the equipment.

5.1.1.4 Accessories. Accessories shall be preserved and packaged method I or III as applicable in accordance with MIL-P-116.

5.1.1.5 Heating element. The heating element supply and return connections shall be sealed with type I, grade A, barrier material conforming to MIL-B-121 and secured with pressure-sensitive tape.

5.1.2 Commercial. The heaters shall be preserved and packaged in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Levels A and B. Packing shall be in accordance with MIL-STD-2073-1. Containers shall be selected from Appendix C, Table VII for the appropriate level.

5.2.2 Commercial. The heaters shall be packed in accordance with ASTM D3951.

5.3 Marking.

5.3.1 Military agencies. Shipments to military agencies shall be marked in accordance with MIL-STD-129.

5.3.2 Civil agencies. Shipments to civil agencies shall be marked in accordance with FED-STD-123.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The radiators covered by this specification are intended for use with hot water and steam heating systems to heat air under natural circulations (thermal convections) conditions.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification
b. Sizes and lengths of radiators required (see 1.2)
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2)
d. When cover, mounting brackets, fasteners are not required (see 3.1)
e. When first article is required for inspection and approval (see 3.2)

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f. Size of steel pipe or tubing to be used for core fabrication (see 3.6.1)
g. Whether core ends shall be threaded or beveled for welding (see 3.6.1)
h. Whether other than rectangular fins are required (see 3.6.2)
i. Treatment and painting, if different (see 3.7)
j. Level of preservation and packing required (see 5.1 and 5.2)

6.3 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.475-1 are invoked and the DD Form 1423 is not used, the data should be delivered by the contractor in accordance with the contract or purchase order requirements.

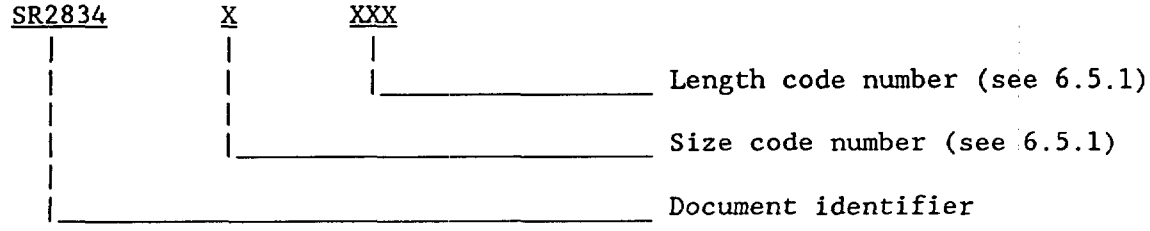
6.3.1 Quality assurance acceptability. The following alternative method should be considered in lieu of the performance test as specified in section 4 when this specification is applied on a contract.

Reference paragraph:

4.5.2 - Alternative evidence of meeting the capacity requirement may be an IBR emblem and applicable data indicating compliance with the IBR rating program or a certified test report from a recognized independent test laboratory indicating the radiator has been tested and conforms to IBR Testing and Rating Code for Finned Tube (commercial) Radiation.

6.4 First article. When a first article inspection is required, the item will be tested and should be a first article sample or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Part or identification number (PIN). A PIN has been established for use to identify the classified item for acquisition (see 1.2). The PIN consists of the document identifier (SR2834) and a PIN code number (see 6.5.1) for the different options for acquisition.



6.5.1 Size and length. The size and length of the radiator (see 1.2) are identified as one-digit and three-digit numbers (see table III).

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TABLE III. Code number to size and length.

Size 1	1	-
Size 2	2	-
Length	-	XXX

XXX is a three-digit number corresponding to the radiator length in inches (see 1.2).

Examples: SR2384-1-024 designates a size 1 (1-1/4-inch (31.75 mm)) nominal pipe size) radiator 2 feet (24 inches (0.61 m)) in length.
SR2384-2-132 designates a size 2 (2-inch (50.8 mm)) nominal pipe size) radiator 11 feet (132 inches (3.35 m)) in length.

6.6 Subject term (keyword) listing.

Finned tube
IBR
Sectional enclosure
Steel fins

6.7 Supersession data. This specification replaces military specification MIL-R-11723E, dated 20 October 1988.

6.8 Classification cross-reference. Classifications used in this specification (see 1.2) are identical to those found in the superseded military specification, MIL-R-11723E.

MILITARY INTERESTS:

Military Coordinating Activity

Navy - YD1

Custodians

Navy - YD1
Air Force - 99

Review Activities

Air Force - 84
DLA - CS

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

PREPARING ACTIVITY:

Navy - YD1

(Project 4520-0366)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	S-R-2834	931230

3. DOCUMENT TITLE
RADIATORS: HEATING, STEEL, MULTIFIN TYPE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
MR. DANNY C. MUI	805-982-5666	551-5666
c. ADDRESS (Include Zip Code) COMMANDING OFFICER, NCBC CODE 1562G 1000 23rd Avenue Port Hueneme, CA 93043-4301	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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